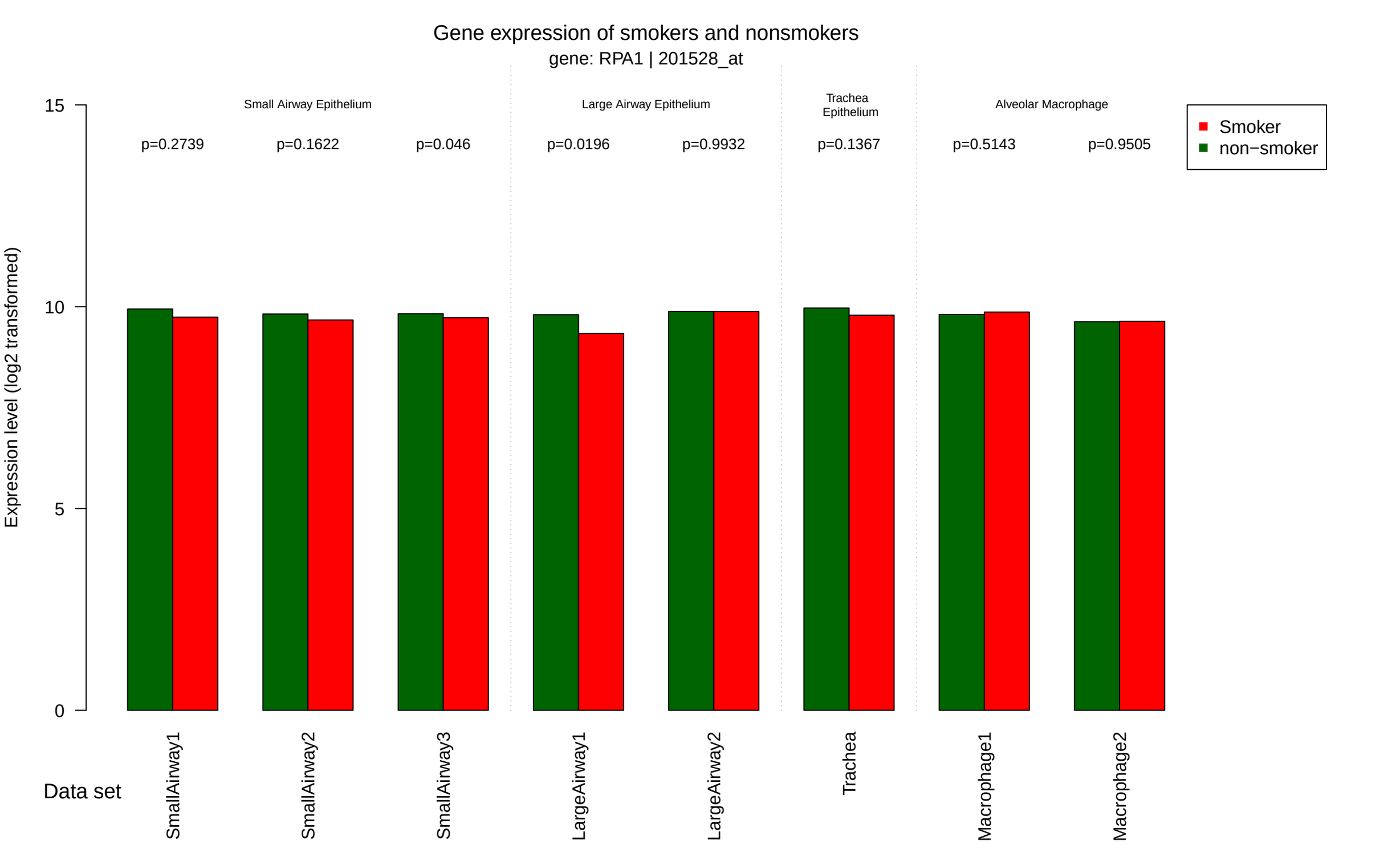
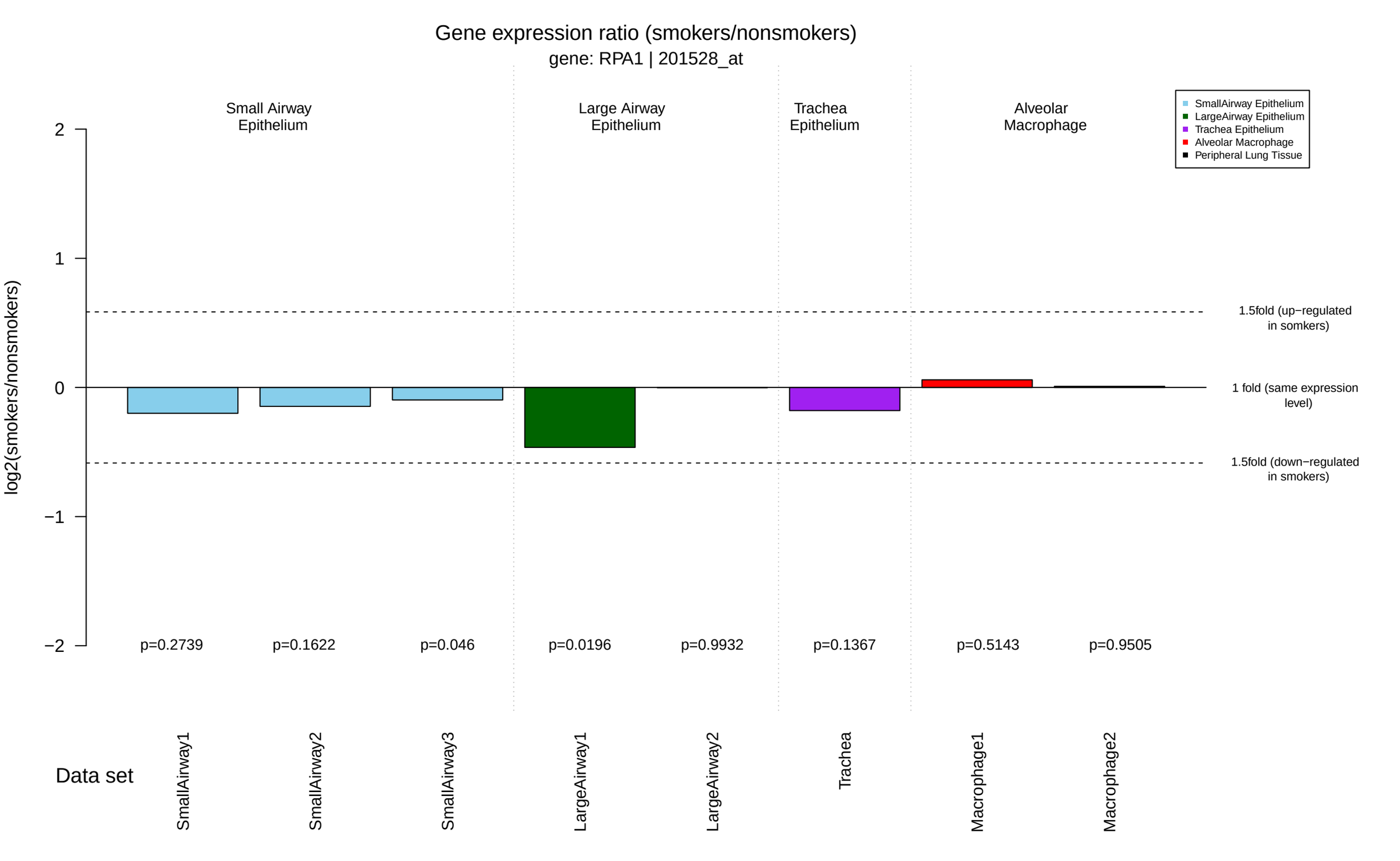
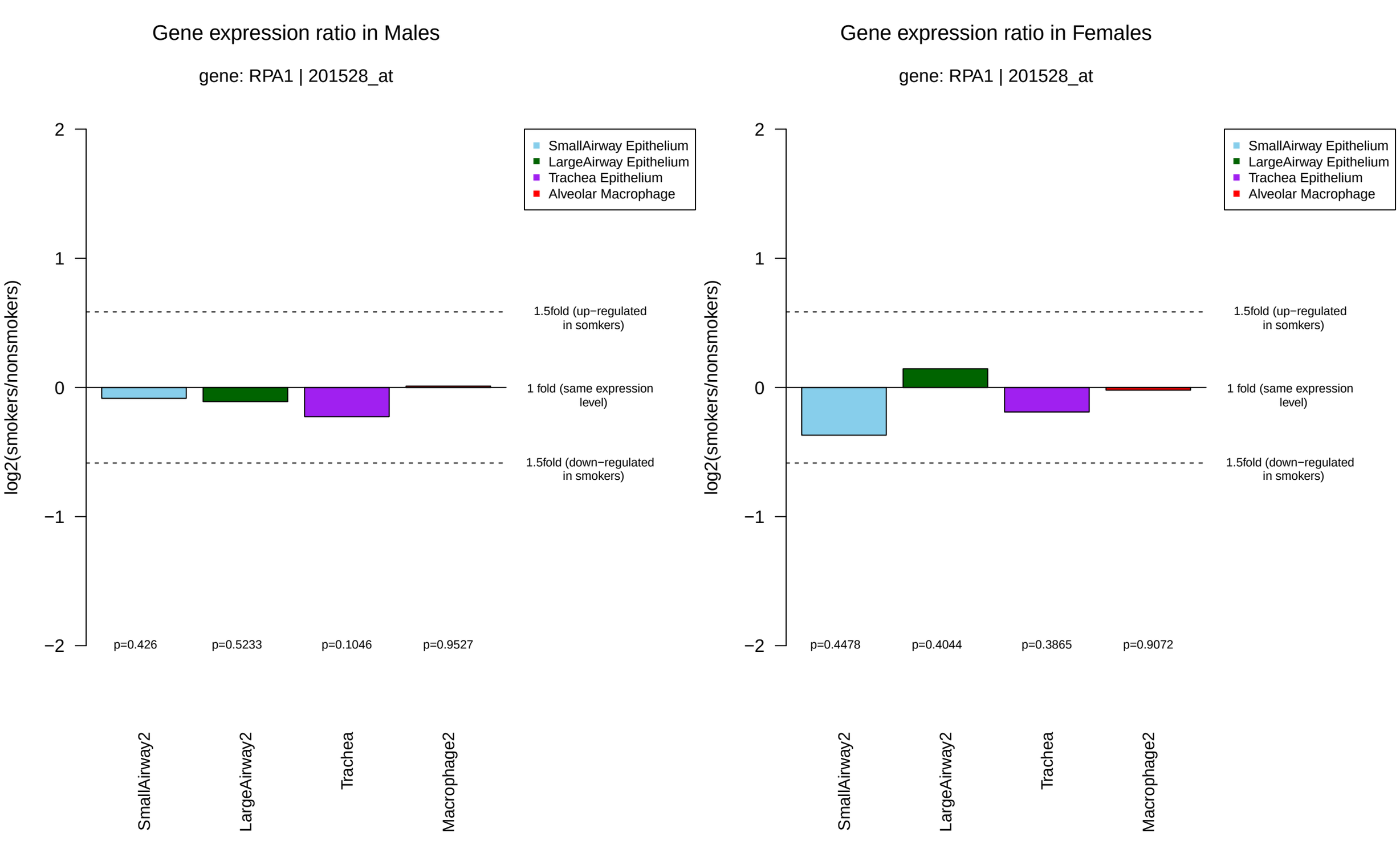
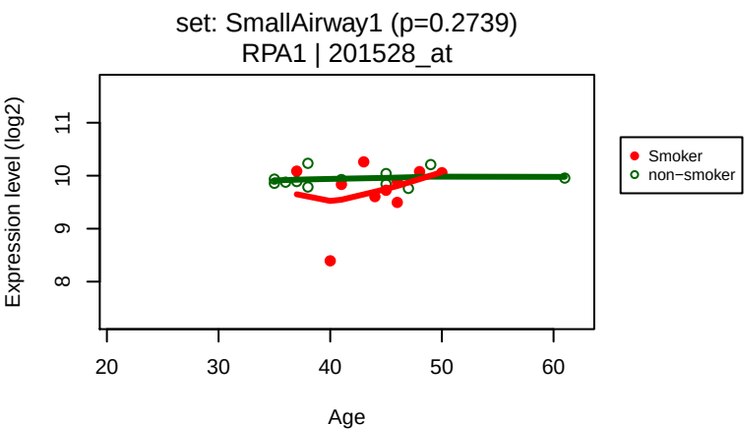
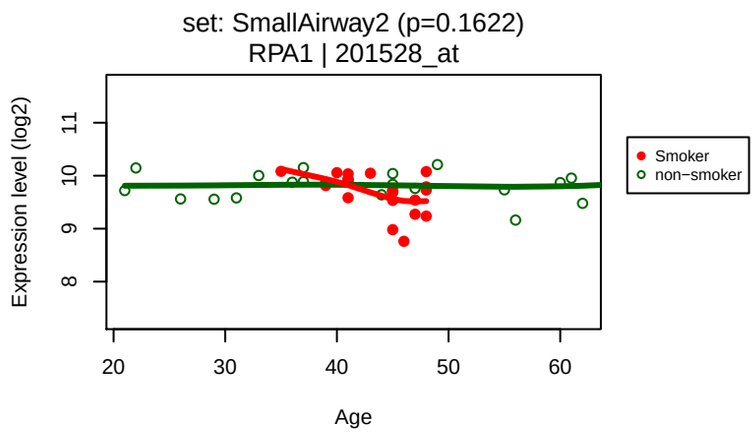
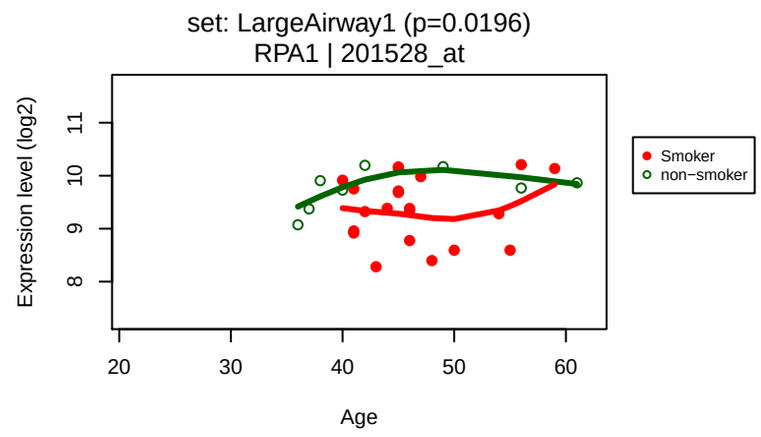

Gene expression of smokers and nonsmokers
gene: RPA1 | 201528_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2739
p=0.1622
p=0.046
p=0.0196
p=0.9932
p=0.1367
p=0.5143
p=0.9505
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

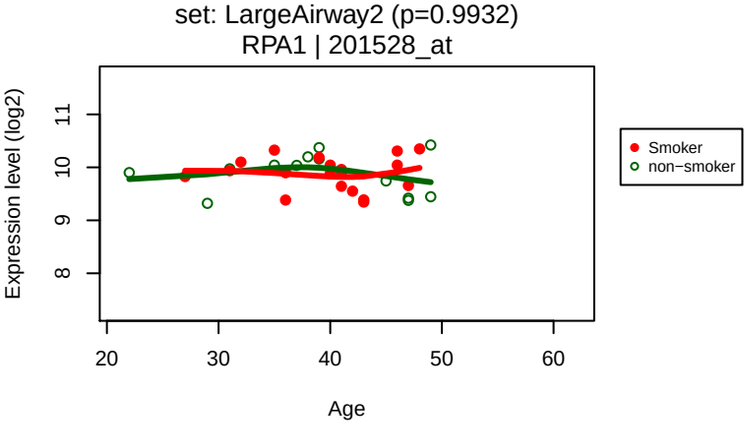

Gene expression ratio (smokers/nonsmokers)
gene: RPA1 | 201528_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2739
p=0.1622
p=0.046
p=0.0196
p=0.9932
p=0.1367
p=0.5143
p=0.9505
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

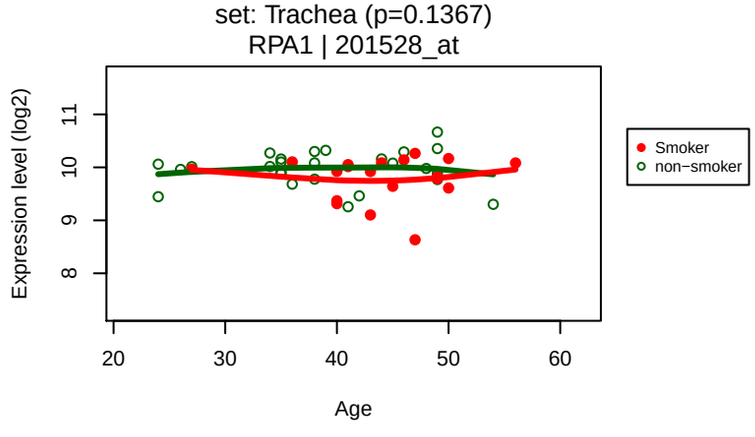

Gene expression ratio in Males
gene: RPA1 | 201528_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.426
p=0.5233
p=0.1046
p=0.9527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPA1 | 201528_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4478
p=0.4044
p=0.3865
p=0.9072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

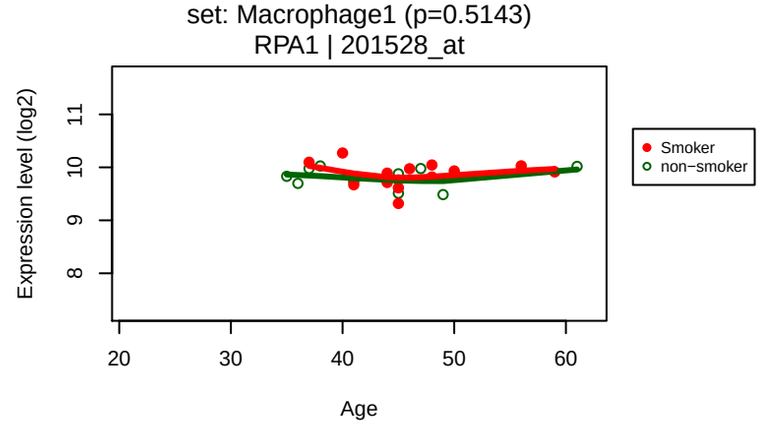

set: SmallAirway1 (p=0.2739)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker

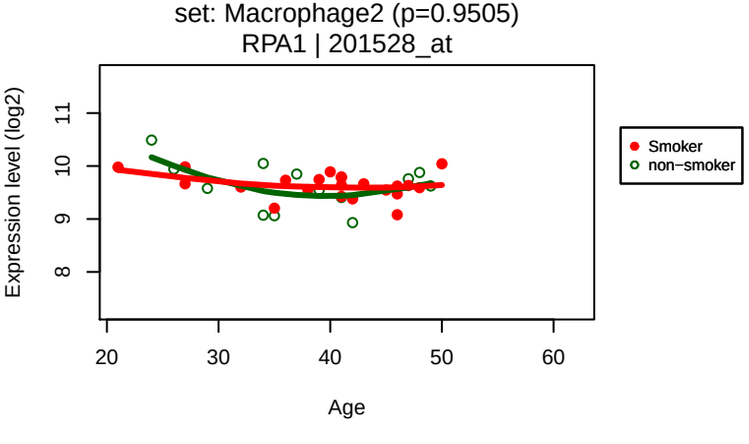

set: SmallAirway2 (p=0.1622)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker

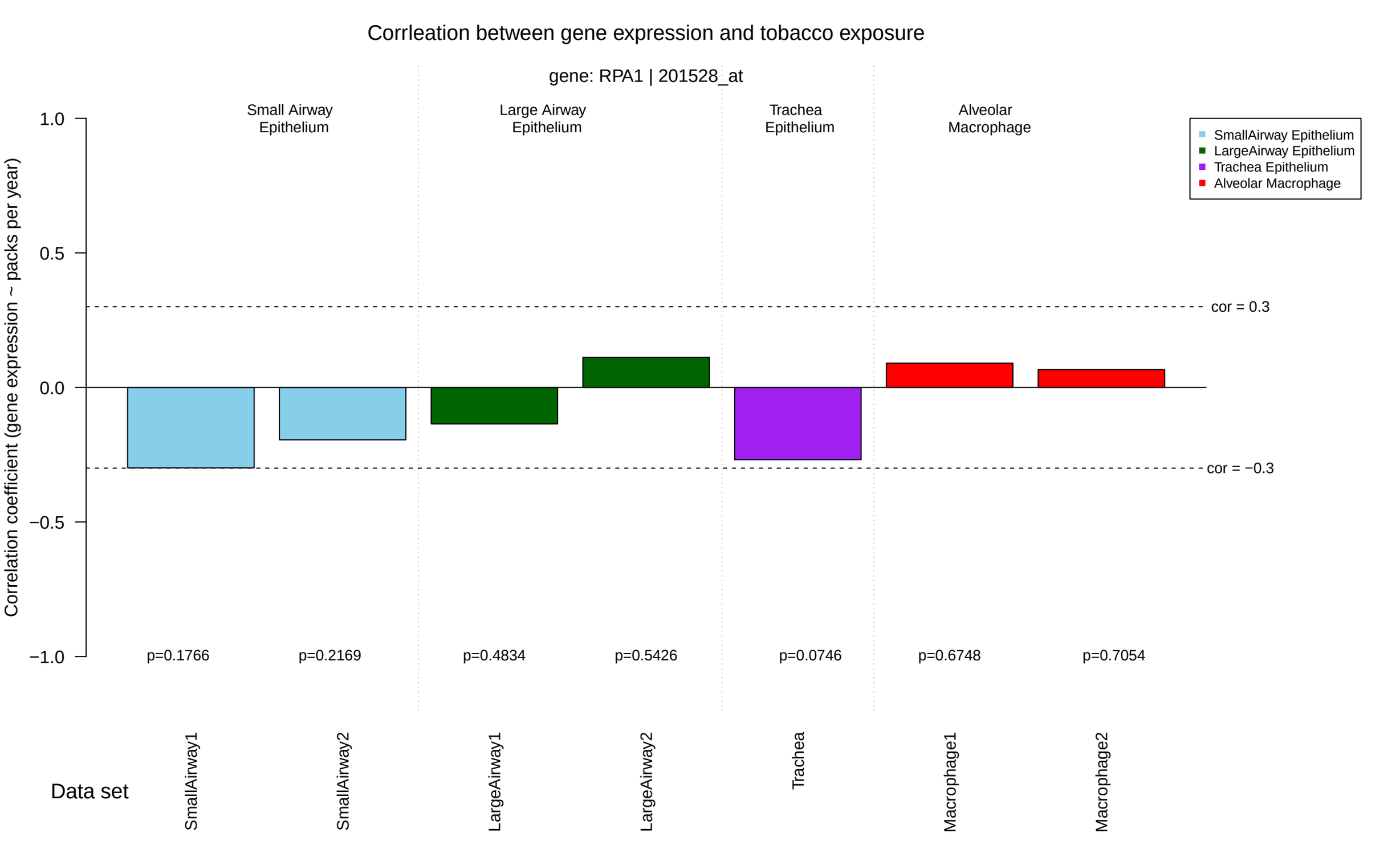

set: LargeAirway1 (p=0.0196)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9932)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1367)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5143)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9505)
RPA1 | 201528_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPA1 | 201528_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1766
p=0.2169
p=0.4834
p=0.5426
p=0.0746
p=0.6748
p=0.7054
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2